

Compiling a C program involves a series of stages to transform the human-readable C source code into machine-executable code. There are typically four main stages in the compilation process:

- **Preprocessing:**
- **Purpose:** The preprocessing stage handles directives, macros, and includes, making the source code more manageable and preparing it for actual compilation.
- **Compilation:**
- **Purpose:** The compilation stage translates the preprocessed code into assembly code or an intermediate representation specific to the target platform.
- **Assembly (Optional):**
- **Purpose:** In some cases, an assembly code generation step is included. It translates the intermediate code into platform-specific assembly language code.
- **Linking:**
- **Purpose:** The linking stage resolves references to external functions and libraries, combining various object files into a single executable file.